5–FLOTATION AND RESERVE BUOYANCY

5.1 FLOTATION AND SURVIVABILITY

A ship at sea is subjected to external loads due to vagaries of weather caused by wind and waves. This results in heavy rolling, pitching and heaving motions. In addition to these, loads may also be generated internally to affect a ship's situation. To survive and remain floating a ship must therefore have adequate buoyancy at all times to combat the constantly changing weather. To achieve this let us first examine the nature of the forces that tend to sink a ship at sea.

5.1.1 Effect of External Downward Forces

Consider a vessel floating initially in equilibrium vertically under the influence of its displacement Δ acting downwards and buoyancy B acting upwards. Under this condition the Principle of Flotation is satisfied and:

$$\Delta = B$$

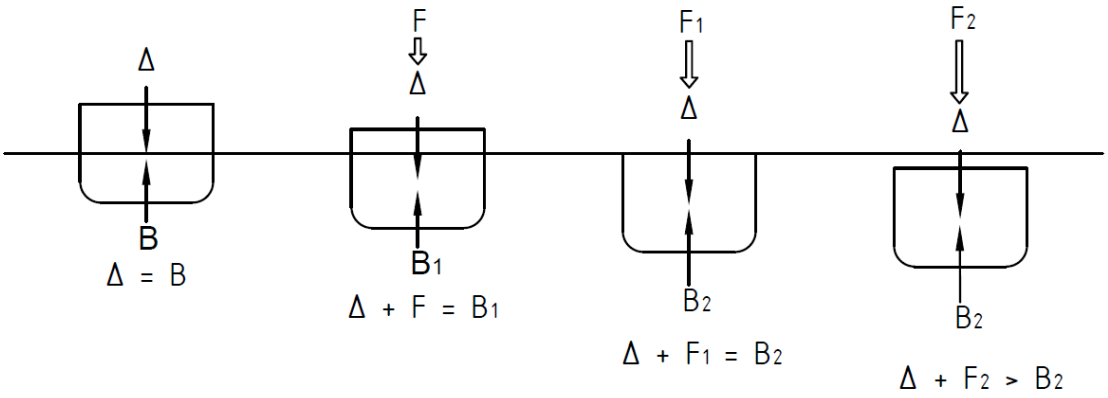


Figure 5.1 - Effect of External Downward Forces

When now the vessel is pushed down by an external force F, for it to continue floating, buoyancy must have to be increased to B_1 to give:

$$\Delta + F = B_1$$

Increase the force to F_1 and for the vessel to remain floating buoyancy must again increase to the maximum B_2 to give:

$$\Delta + F_1 = B_2$$

The vessel is now at the limit of floating, hence if the force should be increased further to F_2 it has a situation where:

$$\Delta + F_2 > B_2$$

At this stage the vessel will sink. It is said to have gone beyond the limit of survivability because the deck has submerged. If therefore a vessel is initially loaded as in the third case to a displacement Δ_1 instead of Δ, the situation is as follows:

$$\Delta_1 = \Delta + F_2 = B_2$$

It will then have no reserve to fall on when external forces act and she will sink on the first impact.

5.1.2 Effect of Symmetrical Flooding

Flooding may be caused through burst piping or breaching of the hull in one way or the other. This apparently results in increase in displacement which if not controlled will result in a behavior similar to that caused by external downward forces illustrated earlier and sinking.

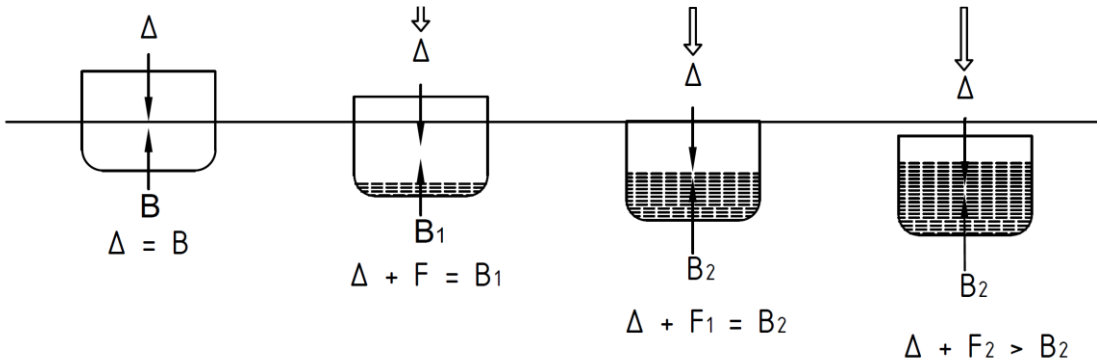


Figure 5.2 - Effect of Symmetrical Flooding

5.1.3 Effect of External Transverse Moments

The actions of external forces may act in the form of moments which tend to roll a ship over to port or to starboard. As soon as the deck edge becomes submerged it will have gone beyond the limit of survivability and will sink. The exact moment that occurs will therefore depend upon the height of deck above water initially, or the reserve buoyancy.

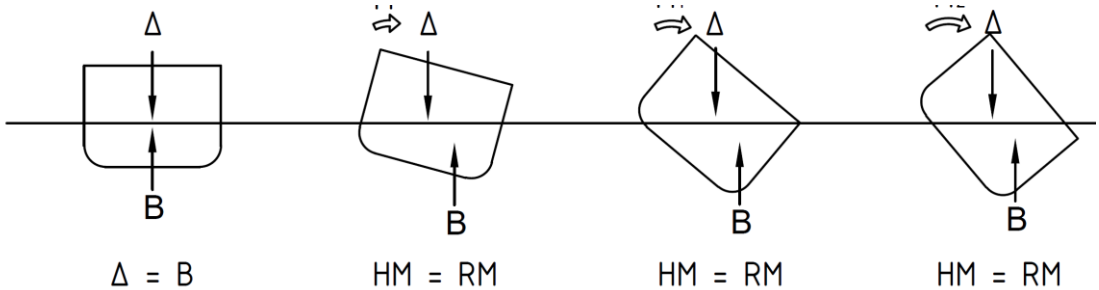


Figure 5.3 - Effect of External Transverse Moments

5.1.4 Effect of Unsymmetrical Flooding

When flooding is not central moments are generated to roll the ship over to port or to starboard. The result in a situation similar to moments caused by external forces.

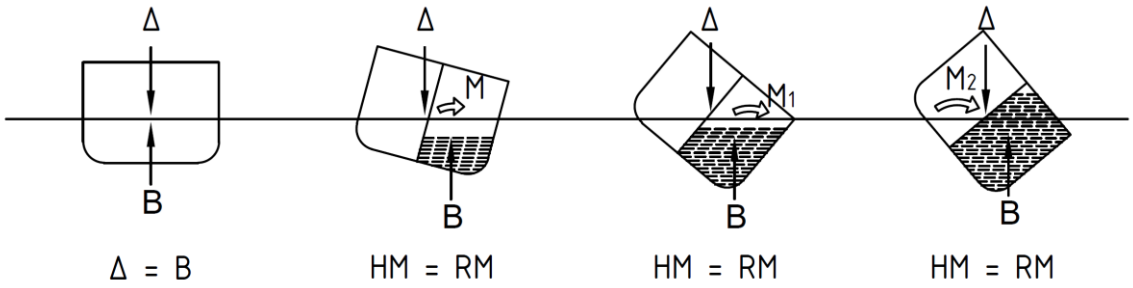


Figure 5.4 - Effect of Unsymmetrical Flooding

5.1.5 Effect of External Longitudinal Moments

This is the situation when moments due to the external forces act longitudinally. The vessel trims seriously to fore or to aft and may go beyond the limit of survivability and sink.

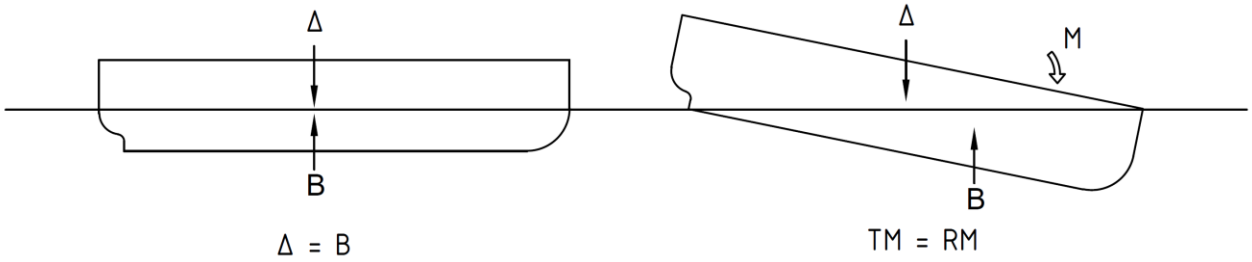


Figure 5.5 - Effect of External Longitudinal Moments

5.1.6 Effect of Fore/Aft Flooding

Similar to longitudinal moments caused by external forces, unsymmetrical flooding in the fore-aft direction may also cause a vessel to be lost if enough buoyancy is not maintained initially.

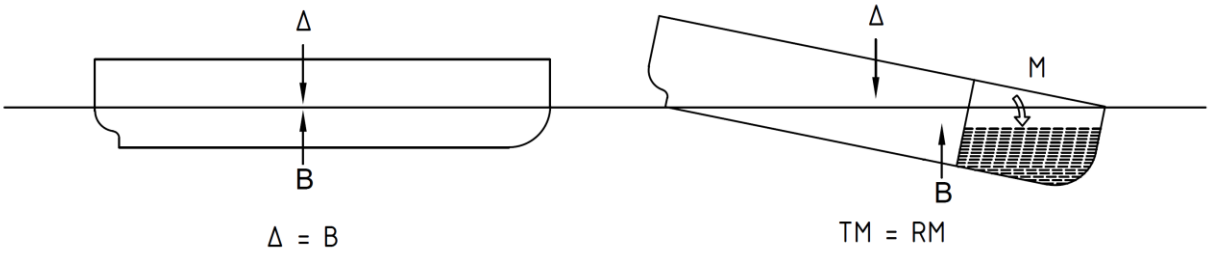


Figure 5.6 - Effect of Fore/Aft Flooding

5.2 FREEBOARD

To prevent the occurrence of any of the situations described in paragraph 6.1 from occurring. It is good practice for ships to maintain reserve buoyancy when fully loaded. It should have been expected therefore that the survival instincts of human beings should be enough to ensure self-regulation when it comes to the maintenance of reserve buoyancy. Unfortunately due to greed that has most of the time not being the case. Especially since ship owning has largely been separated from ship operation by captains and crew. Reserve buoyancy is therefore enforced by law worldwide through the provisions of The International Convention on Load Lines, 1966. It is measured in

terms of Freeboard, i.e. the distance from the deck line at the side to the draught line when a ship is loaded. The markings indicate the minimum freeboard which occurs under fully loaded conditions.

5.2.1 Load Line Marks

The International Convention on Load Lines requires that no ship to which it applies shall proceed to sea on an international voyage unless it has been surveyed, marked and provided with an International Load Line Certificate (1966), or where appropriate, an International Load Line Exemption Certificate.

The load lines and letters shall be painted in white or yellow on a dark background or in black on a light background. They shall also be permanently marked, usually embossed, on both sides of the ship amidships.

Because of wide variations in weather conditions and densities of water worldwide different freeboard markings are used for different waters, appropriately referred to as "zones", in accordance with the following criteria:

Zone	Criteria	How Assigned
S - Summer	Areas with not more than 10 percent winds of force 8 Beaufort (34 knots) or more.	Obtained from Freeboard Table, Reg. 28.
T – Tropical	Areas with not more than 1 percent winds of force 8 Beaufort (34 knots) or more. Not more than one tropical storm in 10 years in an area of 5º square in any one separate calendar month.	$S - \frac{1}{48} \times d_{SM}$
W – Winter	Areas within the Summer zone which seasonally experience weather conditions that are more severe than what usually pertains in the Summer zone.	$S + \frac{1}{48} \times d_{SM}$
WNA – Winter North Atlantic	Special case of Winter zone limited to the North Atlantic area only, for which the seasonal winter weather is longer than the general Atlantic area.	
	For ships of 100 m length or more.	$W + 50\ mm$
	For ships of less than 100 m length	$W + y$ (Where y ≥ 50 mm)
F – Fresh Water	For loading in fresh water in the Summer zone	$S - \frac{\Delta_S}{400TPC_S} mm$
TF – Tropical Fresh Water	For loading in fresh water in the Tropical zone	$T - \frac{\Delta_S}{400TPC_S} mm$

d_{SM} = Summer Moulded Draught
Δ_S = Displacement in Salt Water

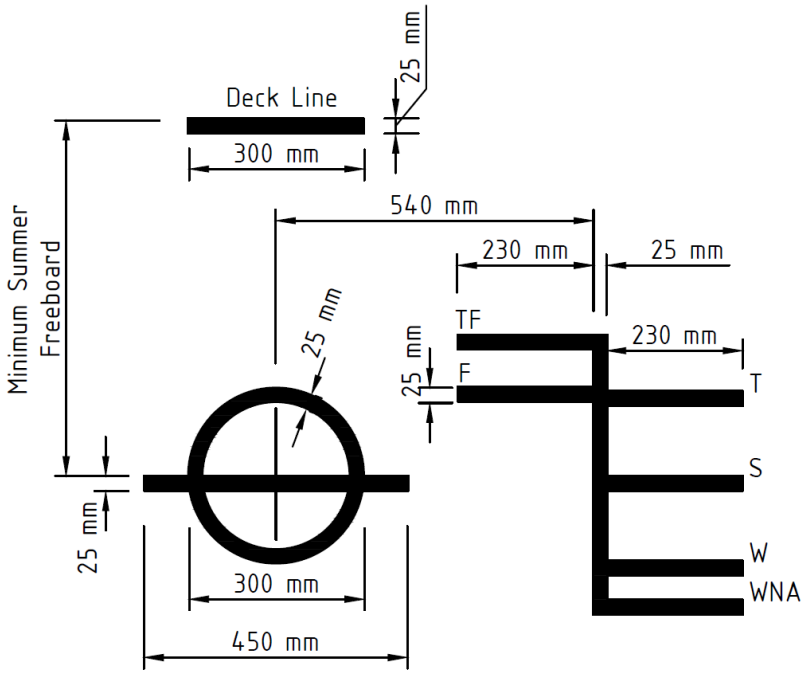


Figure 5.7 – Load Line Mark and lines to be used with this mark

5.3 Sinkage due to Added Mass (Tonne Per Centimetre Immersion)

A useful means of approximating the amounts of load added to or removed from a ship is the use of Tonne Per Centimetre (TPC). TPC is the increase in displacement per unit increase in mean draught.

Consider a vessel initially displacing a mass Δ and volume ∇. By the adding mass (m), its displacement and volume will increase to Δ_1 and volume ∇_1 respectively. This will cause an increase in mean draught or sinkage (s).

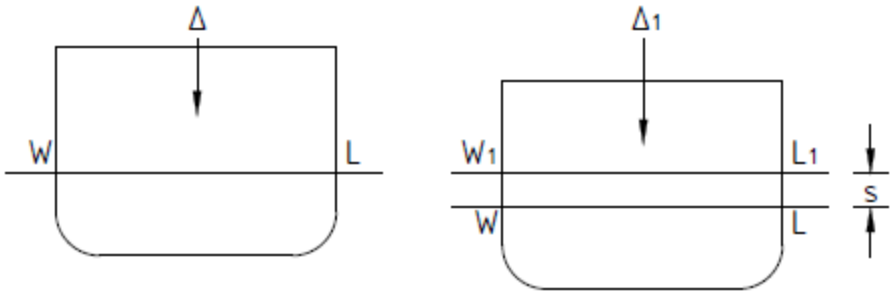


Figure 5.8 – Sinkage due to Added Mass

On assumption that the change in waterplane area from waterline WL to W_1L_1 is negligible, then increase in displacement:

$$\Delta_1 - \Delta = A_W \times s \times \rho$$

In summary, change in displacement of $(\Delta_1 - \Delta)$ tonnes causes mean draught to change by (s) metres.

Therefore for a change in mean draught of one metre, the change in displacement that will be required will be:

$$(A_W \times 1 \times \rho)$$

This is referred to as Tonne per metre immersion:

$$TPM = A_W \rho$$

Or as Tonne Per Centimetre immersion:

$$TPC = \frac{A_W \rho}{100}$$

A curve of TPC plotted against draught will have the same form as the waterplane area curve. TPC may therefore be obtained by extracting from the A_W curve and multiplying the results by the density.

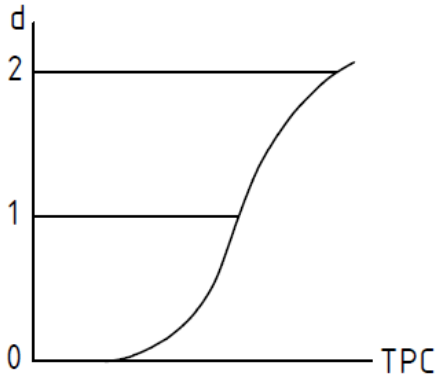


Figure 5.9 – TPC Curve

5.4 Sinkage due to Change in Density (Fresh Water Allowance)

When a ship sails from sea water of density 1.025 tonne/m^3 into fresh water of lower density without change in displacement, then as a result of the difference in densities there will be change in volume of displacement.

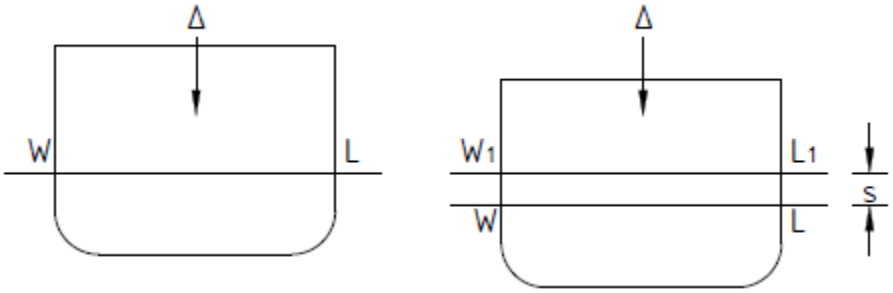


Figure 5.10 – Sinkage due to Change in Density

Assuming the A_W remains constant and representing densities of sea water and fresh water by (ρ_S) and (ρ_F) respectively, then increase in volume of displacement:

$$\nabla_F - \nabla_S = A_W \times S$$

$$s = \frac{\nabla_F - \nabla_S}{A_W} = \frac{\frac{\Delta}{\rho_F} - \frac{\Delta}{\rho_S}}{A_W}$$

Therefore

$$s = \frac{\Delta}{A_W}\left(\frac{1}{\rho_F} - \frac{1}{\rho_S}\right) = \frac{\Delta}{A_W}\left(\frac{\rho_S - \rho_F}{\rho_F \rho_S}\right) \text{ m}$$

Or

$$s = \frac{100\Delta}{A_W}\left(\frac{\rho_S - \rho_F}{\rho_F \rho_S}\right) \text{ cm}$$

But

$$\frac{100}{A_W \rho_S} = \frac{1}{TPC_S}$$

Therefore

$$s = \frac{\Delta}{TPC_S}\left(\frac{\rho_S - \rho_F}{\rho_F}\right) \text{ cm}$$

For the special case of sailing into pure fresh water of ρ_F = 1.000 tonne/m^3

$$s = \frac{\Delta}{TPC_S}\left(\frac{1.025 - 1.000}{1.000}\right) = \frac{\Delta}{TPC_S}\left(\frac{0.025}{1.000}\right) = \frac{\Delta}{40TPC_S} \text{ cm}$$

This change in density is referred to as the Fresh Water Allowance (FWA). i.e.:

$$FWA = \frac{\Delta}{40TPC_S} \text{ cm}$$

5.5 Deadweight Scale

The deadweight scale is as drawing that provides a method for estimating the additional draft or for determining the extra load that can be taken onboard when a vessel is being loaded in water of density less than that of salt water. The following scale displays readings for freeboard, deadweight in salt water and in fresh water, draft of the ship (mean), displacement in salt water and in fresh water, tonnes per cm (TPC) in salt water and in fresh water, and moment to change trim 1 cm (MCTC).

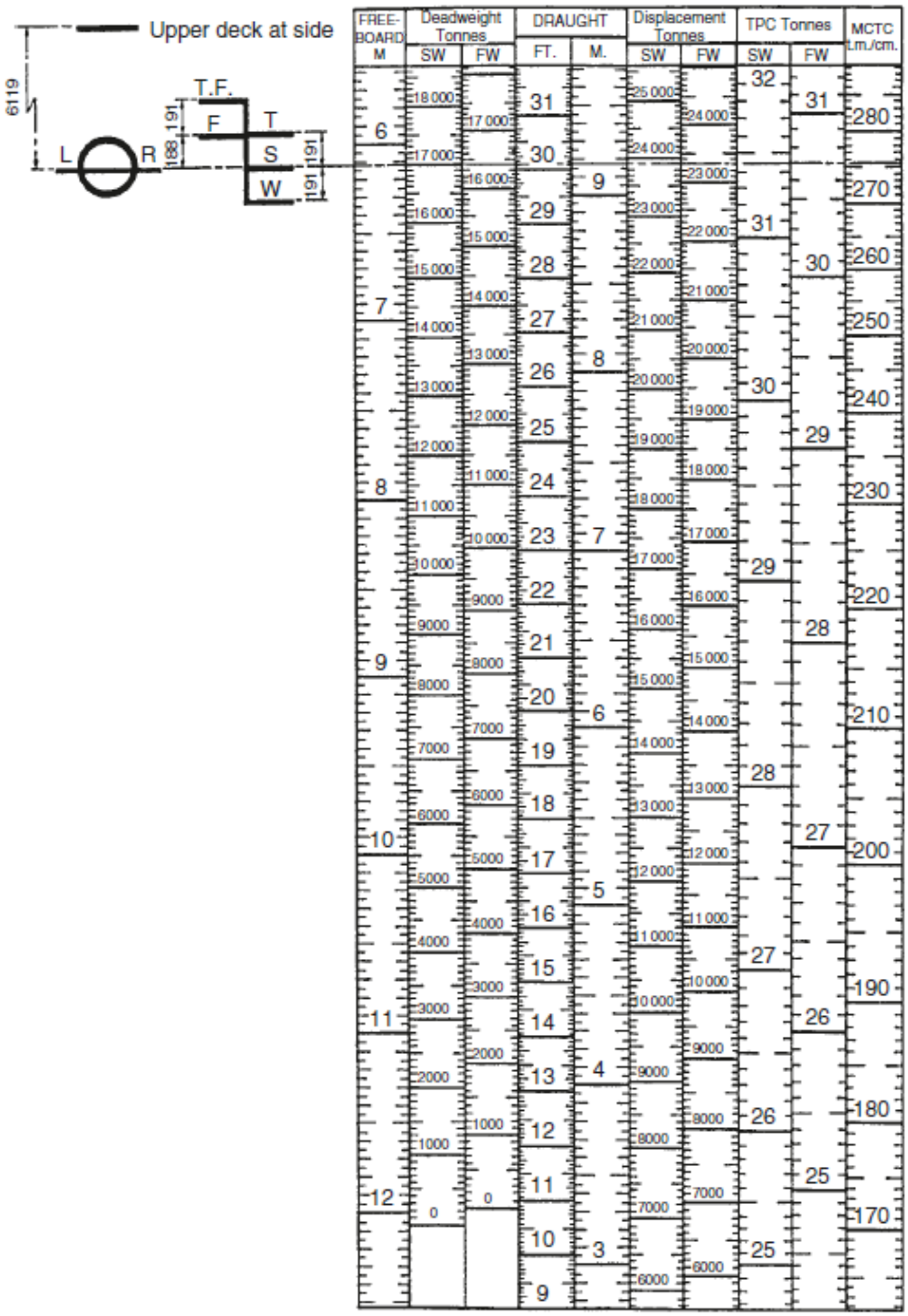


Figure 5.11 - Deadweight Scale

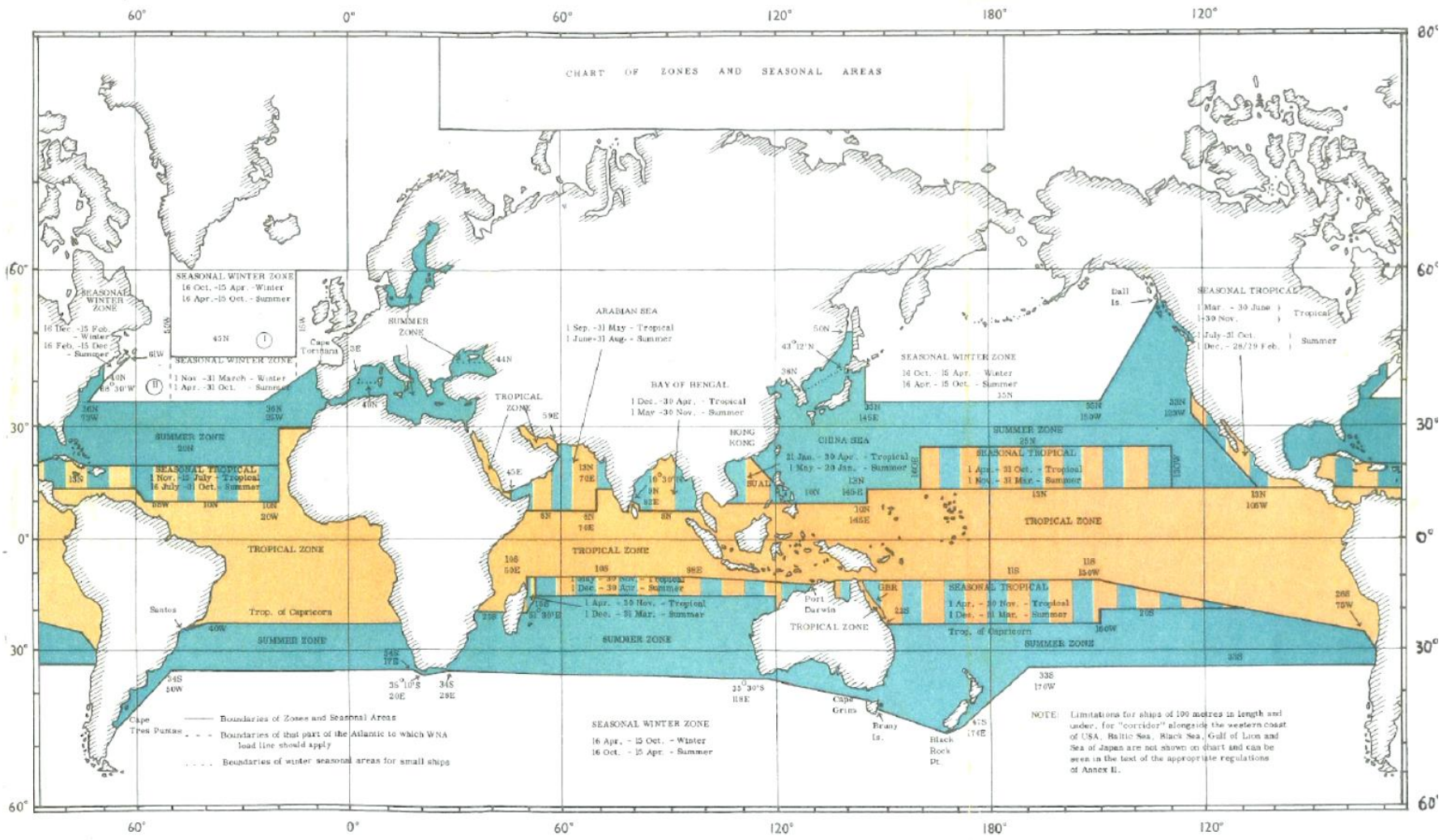


Figure 5.12 – Chart of Zones and Seasonal Areas